

POS system for Coffee Shop

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CECS 491A

Staff (Barista) Use Cases

1. Login/Logout — Authenticate with credentials, auto-timeout on inactivity

Use Case #1	Login/Logout	
Actor	Staff (Barista)	
Preconditions	● Staff must have a registered account ● Staff must have a network connection ● Staff is not already logged in (for login). ● Staff is logged in (for logout).	
Flow of Control (Login)	Enter username and password	
		Validates entered credentials
	Clicks the “Login” button	
		Checks account status If valid, it creates a user session Displays the dashboard screen
Flow of Control (Logout)	Staff clicks “Logout.”	

		Displays the logout confirmation dialog
	Confirms logout	
		Terminates active session Redirects the user to the login screen
Post Conditions	● Staff is successfully logged in or logged out ● A session is created or terminated	
Error Conditions	● Invalid username or password ● Network connection lost ● The account is deactivated ● Session timeout due to inactivity	
Non-functional requirements	● Login must be completed within 3 seconds ● Auto-timeout after 10 minutes of inactivity	

2. **Create Order** — Select items, customize (size, milk, add-ons), add to order

Use Case #2	Create Order	
Actor	Staff (Barista)	
Preconditions	● Staff must be logged in ● Menu items must exist ● The system must	

	have a network/database connection	
Flow Of Control	Selects “New Order”	
		Displays available menu items
	Selects one or more items	
		Adds selected item(s) to the temporary order list
	Customizes item (size, milk type, add-ons)	
		Updates item details and recalculates price
	Adds items to the order	
		Displays updated order summary
	Reviews order summary	
		Shows subtotal and total price
	Confirms order	
		Saves order in the database Sets order status to “Pending.” Displays a confirmation message
Post Conditions	● The order is saved in the database ● Order status is set to “Pending.” ● The order appears in the active orders list	

Error Conditions	● Item unavailable ● Invalid customization ● Network/database failure	
Non-functional requirements	● Order creation must be completed within 5 seconds ● The system must handle multiple simultaneous orders	

3. **Modify Order** — Edit items/customizations before payment is processed

Use Case #3	Modify Order	
Actor	Staff (Barista)	
Preconditions	● Staff must be logged in ● Order must exist ● Payment must not yet be processed	
Flow of Control	Selects an existing order from the active orders list	
		Displays order details and item breakdown
	Edits items, quantities, or customizations	
		Updates order details in the temporary view
	Removes or adds items as needed	
		Recalculates the subtotal and total price

	Clicks "Save Changes"	
		Validates modifications Displays a confirmation message Updates order information in the database
Post Conditions	● Order details are updated in the database ● The updated order is displayed	
Error Conditions	● Order not found ● Network connection lost	
Non-functional requirements	● Modification must be completed within 3 seconds	

4. **Process Payment** — Accept cash or electronic payment, receive confirmation

Use Case #4	Process Payment	
Actor	Staff (Barista)	
Preconditions	● Staff must be logged in ● Order must exist ● The order must not be paid for	
Flow of Control	Selects "Process Payment" for an order	
		Displays available payment options (cash, credit/debit, mobile payment)

	Selects payment type	
		Prompts for payment details (if electronic) or cash amount
	Enter payment information or confirm cash received	
		Validates payment details
	Confirms payment	
		Processes transaction through payment gateway (if electronic) Records payment in the database Updates order status to "Completed." Generates and displays a receipt Displays payment confirmation message
Post Conditions	● Payment recorded in the database ● Order status updated to "Completed" ● Receipt generated	
Error Conditions	● Payment declined ● Insufficient funds ● Network or payment gateway failure	
Non-functional requirements	● Payment processing must be completed within 5 seconds ● Transaction data must be securely	

	encrypted	
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5. Update Order Status — Move order through Pending → In Progress → Completed

Use Case #5	Update Order Status	
Actor	Staff (Barista)	
Preconditions	● Staff must be logged in ● Order must exist	
Flow of Control	Staff selects the order	
		Displays current order details and status
	Staff updates status (Pending -> In Progress -> Completed)	
		Validates status transition
	Confirms status update	
		● Saves updated status in the database ● Refreshes order display ● Updates the order in the active orders queue ● Displays a confirmation message
Post Conditions	● Order status updated in the database ● Status visible to relevant users	
Error Conditions	● Invalid status	

	transition ● Network failure	
Non-functional requirements	● Status update must be completed within 2 seconds	

6. **View Menu** — Browse current menu items, prices, and availability

Use case # 6	Staff's Menu	
Actor	Staff	
Pre-condition	● Staff is logged into the system and has appropriate permissions ● Allow staff to view the current menu, including prices, availability status, and operational details (item status, hidden items, stock levels). ● System has stable connection	
Flow of control		
	Staff logs into the system.	
		The system authenticates credentials and grants access based on role.
	Staff selects View Menu from dashboard	
		The system retrieves and displays the current menu list grouped by categories.
	Staff selects a menu item	

		The system displays detailed information (price, availability, stock level, status: active/inactive).
	Staff reviews availability(items marked out of stock)	
		The system shows real-time status and highlights unavailable items.
	Staff refreshes or navigates back	
		System updates menu data if needed and returns to the previous screen.
Post-condition	● Staff successfully views the current menu status. No changes are made unless another use case is triggered.	
Error- condition	● Incorrect role permissions exposing restricted data ● Inventory synchronization delays ● Performance issues with large menu	
Non-functional requirements	● The system must allow staff logged in with valid credentials. ● Menu items display name, category, price, and availability with less than 3s ● The menu reflects real-time updates.	

7. **View Active Orders Queue** — Real-Time List of Pending Orders/Displays order details and next actions

Use Case #7	View Active Orders Queue	
Actor	Staff (Barista)	
Preconditions	● The staff user is logged into the POS system. ● The system database is operational. ● There are existing orders in Pending or In Progress state. ● The user has permission to access the order queue. ● Internet/network connection is active.	
Flow of Control	Barista selects “Active Orders” from the dashboard.	
		System queries database for all orders with status = Pending or In Progress.
	Barista views displayed order queue.	
		System displays orders sorted by timestamp (FIFO order).
	Barista selects a specific order.	
		System displays order details: Order ID, items, customizations, payment status, timestamp.

	Barista chooses to update order status	
		System updates order status in database.
	Barista marks the order as "Completed."	
		System removes order from Active Queue and logs completion time.
	Barista refreshes page.	
		System reloads and syncs queue in real-time without duplicating records.
Post Conditions	● Active order list reflects real-time database state. ● Completed orders are removed from queue. ● Updated status is saved permanently. ● Order details are accurately displayed.	
Error Conditions	● If database fails → Display error message: "Unable to retrieve active orders." ● If no active orders → Display message: "No active orders available." ● If update fails →	

	Display: "Order update unsuccessful. Try again." ● If network lost → Show offline indicator and disable updates. ● If unauthorized access → Display access denied message.	
Non-functional requirements	● Real-time refresh must occur within 2 seconds. ● The system must handle a minimum 100 concurrent active orders. ● The interface must be optimized for tablet POS screens. ● Queues must remain responsive under peak hours. ● System uptime: 99% during business hours.	

Admin (Owner/Manager) Use Cases

8. **Manage Menu Items** — Add, edit, remove items (name, description, category, price, availability)

Use case # 8	Owner's menu
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Actor	Owner/admin	
Pre-condition	● Admin authentication ● System has stable connection	
Flow of control	Admin navigates to Menu and selects Add Item	
		● The system validates the user to ensure the user is authorized admin. ● The system presents an item creation form with a field: items, price, description, item available.
	Admin fills out the form	
		● The system validates the input. ● The system displays user's input
	Admin previews the item,	

	click save if agree or modified if not	
		● If the admin agrees what is displayed, the system creates the items record. ● The system pushes the update to live menu ● If the admin does not agree, the system allows the admin to modify the data.
	Admin receives the confirmation after reviewing	
		System displays success message and ask if the admin want to log out
	Admin log out	
		System log out successfully

Post-condition	Items are created/updated/deactivated in the menu database; audit trail recorded; live menu reflects changes once published/synced
Error- condition	● If required fields are missing or invalid (e.g., negative price), system rejects save and displays field-specific errors. User corrects and retries. ● If user lacks permission, system denies access and logs attempt. ● If another admin edited before saving, system warns and shows different admin can merge, discard, or force overwrite. ● If publish to downstream systems fails, system saves locally, marks update as pending and retries
Non-functional requirements	● Item create/update actions should complete within 2 seconds for single items. ● All actions require strong authentication ● Keep immutable audit logs

9. **Manage Inventory** — View stock levels, update quantities, set low-stock thresholds

Use Case #9	Manage Inventory
Actors	Admin (Owner/Manager)

Pre-Conditions	● Admin must be logged in ● Inventory items must exist in database ● Network/Database connection must be available	
Flow of Control	Admin opens “Inventory” dashboard	
		System displays current stock levels + low-stock thresholds
	Admin updates quantities or thresholds	
		System validates values (non-negative, reasonable bounds)
	Admin presses “Save”	
		System commits updates and syncs changes in real time
Post Conditions	● Inventory levels/thresholds updated and stored ● Updated values affect order availability rules	
Error Condition	● Invalid quantity/threshold ● Inventory record not found ● Network/Database failure to connect	
Non-Functional Requirements	● Inventory updates visible across system in real time ● Audit trail recorded for inventory changes (recommended for integrity)	

10. **View Sales Reports** — Daily/weekly/monthly summaries, revenue analytics, peak times

Use Case #10	View Sales Report	
Actor	Admin (Owner/Manager)	
Pre-Conditions	● Admin must be logged in ● Sales/Transaction data exists ● Network/Database connection is available	
Flow of Control	Admin opens “Reports” dashboard	
		System shows report types (Daily/Weekly/Monthly, revenue, best-sellers, peak times, etc.)
	Admin selects date range/report type	
		System aggregates sales metrics and displays charts/tables
	Admin drills into details (selects a specific metric, date, or product)	

		System displays detailed transaction list and item-level breakdown
Post Conditions	● Admin sees requested report and metrics details	
Error Condition	● No data for selected range ● Report generation failure ● Network/Database failure to connect	
Non-Functional Requirements	● Reports load within 5 seconds for typical date ranges ● Access restricted to Admin role (no staff access)	

11. Export Reports — Download reports as CSV or PDF

Use Case #11	Export Reports	
Actor	Admin (Owner/Manager)	
Pre-Conditions	● Admin must be logged in ● A report is generated/available to export	
Flow of Control	Admin selects “Export” on a report	
		System shows export formats (CSV/PDF)

	Admin chooses format	
		System generates file
	Admin downloads file	
		System provides downloadable file and confirms export success
Post Conditions	● Export file generated and delivered to admin in (CSV/PDF) format	
Error Condition	● Export generation fails ● Network error during download ● Network/Database failure to connect	
Non-Functional Requirements	● Export completes within 10 seconds for typical reports ● Exported files match on-screen data (consistency)	

12. Manage Staff Accounts — Create, edit, deactivate staff users and assign roles

Use Case #12	Manage Staff Accounts	
Actor	Admin (Owner/Manager)	
Pre-Conditions	● Admin must be logged in ● Admin has user-management permissions ● Network/Database connection is available	

Flow of Control	Admin opens “Staff Accounts”	
		System displays list of staff users and roles
	Admin selects Create, Edit, or Deactivate	
		System displays account form
	Admin enters or modifies user information	
		System validates input and role assignments
	Admin clicks “Save”	
		● System updates user record ● System logs action ● System displays confirmation message
Post Conditions	● Staff account created, updated, or deactivated ● Role permissions applied immediately ● Audit trail recorded	
Error Condition	● Duplicate username/email (if enforced) ● Invalid role assignment ● Network/Database failure to connect	
Non-Functional Requirements	● RBAC must be enforced at all	

	times ● Authentication and account changes must be logged ● Changes must complete within 3 seconds	
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13. **Process Refund** — Issue refunds on completed transactions

Use Case #13	Process Refund	
Actor	Admin (Owner/Manager)	
Pre-Conditions	● Admin must be logged in ● Transaction exists and is eligible for refund ● Network/Databas e/Payment system connection is available (E-Refunds)	
Flow of Control	Admin opens “Transaction History”	
		System displays searchable transaction list
	Admin selects a completed transaction	
		System displays transaction details and refund option

	Admin selects refund type (Full/Partial)	
		System validates refund eligibility
	Admin confirms refund	
		● System processes refund through payment gateway ● System updates transaction record ● System logs refund action ● System displays confirmation message
Post Conditions	● Refund recorded with timestamp ● Transaction marked Refunded or Partially Refunded ● Audit record stored	
Error Condition	● Transaction not found ● Refund not permitted ● Payment gateway failure ● Network/Databas e failure	
Non-Functional Requirements	● Refund processing must be secure and encrypted ● Refund actions must be logged ● Refund must complete within 5 seconds	

14. **View Transaction Logs** — Audit detailed payment/order history

Use Case #14	View Transaction Logs	
Actor	Admin (Owner/Manager)	
Pre-Conditions	● Admin must be logged in ● Transaction log data exists ● Network/Databas e connection is available	
Flow of Control	Admin opens “Transaction Logs”	
		System displays searchable log list (date, amount, method, order reference)
	Admin filters or searches logs	
		System refreshes list with matching results
	Admin selects a transaction	
		System displays full transaction details
Post Conditions	● Admin can audit transactions and view details	
Error Condition	● No results found	

	for selected filter ● Network/Databas e failure ● Unauthorized access attempt	
Non-Functional Requirements	● Logs must load within 3–5 seconds ● Access restricted to Admin role ● Log data must be tamper-resistant	